

Nicholas Fitzpatrick

nicholasfitzpatrick@berkeley.edu | San Francisco Bay Area, CA | linkedin.com/in/n-fitzpatrick

UC Berkeley Industrial Engineering and Operations Research Master's graduate (May 2026) with hands-on experience building and deploying GenAI systems, retrieval-augmented architectures, and LLM-based workflows. Focused on roles where AI drives measurable operational and business outcomes.

EDUCATION

University of California, Berkeley – College of Engineering

Aug 2025 – May 2026

Master of Analytics – Industrial Engineering and Operations Research, Berkeley, CA

- Coursework: Deep Learning (CNNs, Transformers, LLMs), Machine Learning, Supply Chain Analytics (demand forecasting, facility location, inventory optimization), Optimization, Simulation, Data Engineering, Database Systems
- IEOR Department scholarship for exceptional merit and character

University of California, Berkeley – College of Environmental Design

Jun 2018 – May 2020

Bachelor of Arts in Sustainable Environmental Design – with Honors, Berkeley, CA

- Focused on geospatial and energy data analysis as applied to architecture and urban planning
- Lectured as a Student Instructor for UC Berkeley School of Public Health on energy, climate, and governance

Santa Rosa Junior College

Aug 2023 – May 2025

A.S. Computer Studies – Web Full-Stack Developer | A.A. Computer Studies – Web and Multimedia, Santa Rosa, CA

PROFESSIONAL EXPERIENCE

Geopogo – Berkeley, CA

Oct 2025 – Present

AI Engineer & Product Lead(Internship)

- Built and deployed an end-to-end pipeline converting architectural drawings into labeled training sets for a building code compliance model; onboarded external architecture firm partners for live compliance testing and production inference, expanding training coverage through active data acquisition.
- Integrated Google GenAI capabilities for image generation and generative design workflows via Python and JavaScript; developed frontend features and led UX/UI decisions to streamline architect and designer workflows across the platform.
- Coordinated a cross-functional team of 10 across engineering and product to integrate the data pipeline and compliance model into the core platform; managed iterative delivery from user feedback through to shipped features.
- Shipped redesigned landing page and consolidated site navigation, improving user onboarding experience for architecture and design firm clients.

Outlier – Remote

Oct 2022 – May 2024

AI Data Specialist (Contract)

- Applied RLHF and hallucination-control evaluation to LLM outputs across agentic and multi-step reasoning tasks, improving accuracy and safety adherence in production-grade systems.
- Conducted prompt engineering experiments and performed quality assurance on model responses and peer evaluations.

Cook's Vineyard Management – Sonoma, CA

Sept 2020 – Oct 2022

Operations Manager

- Owned end-to-end operations across 475 acres and 15+ clients, managing dynamic resource allocation, real-time exception handling across weather and equipment constraints, and crew routing under seasonal SLAs.
- Built a spreadsheet-based scheduling and monitoring system to track crew assignments, equipment utilization, and job completion across concurrent sites; reduced idle time and rework via daily reallocation logic.

PROJECTS

RAID – Retrieval-Augmented Inverse Dynamics for Robotic Manipulation

Spring 2026

INDENG 242B – Deep Learning

- Designed and implemented a retrieval-augmented architecture that solves a core robotics challenge: given a world model's predicted future visual state, infer the motor command to reach it, combining a learned network with a cross-attention retrieval mechanism over a demonstration memory bank.
- Achieved a 6.5x reduction in action prediction error vs. a direct neural network baseline on LIBERO-Spatial with only 25 demonstrations (val MSE 0.131 vs. 0.852); retrieval advantage persisted through 200 demonstrations, validating retrieval-augmented systems as sample-efficient alternatives to fully learned models.
- Extended the architecture to reinforcement learning fine-tuning in simulation, reaching 25% task success and demonstrating that retrieval-augmented policies can be improved through online interaction without retraining from scratch.

LEO Ground-Station Network Design

Spring 2026

INDENG 230 – Supply Chain Analytics

- Modeled LEO satellite ground-station placement as a supply chain network design problem, formulating a mixed-integer linear program to trace cost-coverage Pareto frontiers across competing infrastructure constraints, including backhaul availability and geopolitical feasibility.
- Key finding: demand model selection is a structural capital allocation decision. Population-weighted coverage reached 43.3% vs. 20.8% uniform at identical geometry; adding geopolitical constraints reduced median distance to actual Starlink gateways from 7,154 km to 881 km, validating the model against real-world infrastructure.

Understanding Customer Churn for BizGrow

Fall 2025

Google Hackathon

- Led a team to analyze SaaS customer journey data using Python and Pandas, applying keyword clustering and segmentation to isolate a high-revenue cohort responsible for ~80% of churn revenue loss; validated findings with Causal ML models (Double ML, Uplift) to rule out confounding variables.
- Delivered an executive-ready presentation translating churn analysis into a prioritized action plan addressing \$29M in at-risk contracts.

VOLUNTEERISM

California Academy of Sciences – San Francisco, CA

Dec 2022 – May 2023

Scientific Computing Volunteer

- Built a Flask/Python dashboard with Bash collection agents to monitor CPU/GPU usage, job queues, and storage across research clusters; informed equitable compute scheduling for the Director of Scientific Computing.

SKILLS & INTERESTS

Languages: Python, SQL, TypeScript, JavaScript, C++, HTML/CSS

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, LangChain, FAISS, Sentence-Transformers, Spark

Platforms & Tools: Claude Code, Google Cloud SQL, Databricks, Firebase, MongoDB, Git, Dash/Plotly

Methods: RAG, Embedding-Based Search, LLM Agents, LLM Fine-Tuning, MLOps, Model Deployment & Versioning, Prompt Engineering, Multi-step Reasoning, Hallucination Control, Vector Search, Statistical Modeling, ETL, REST APIs, A/B Testing

Other: Instructing & Public Speaking, Cross-functional Team Leadership, GIS, UX/UI Design

Interests: Scuba Diving, Motorcycling, Photography, Aquarium Design